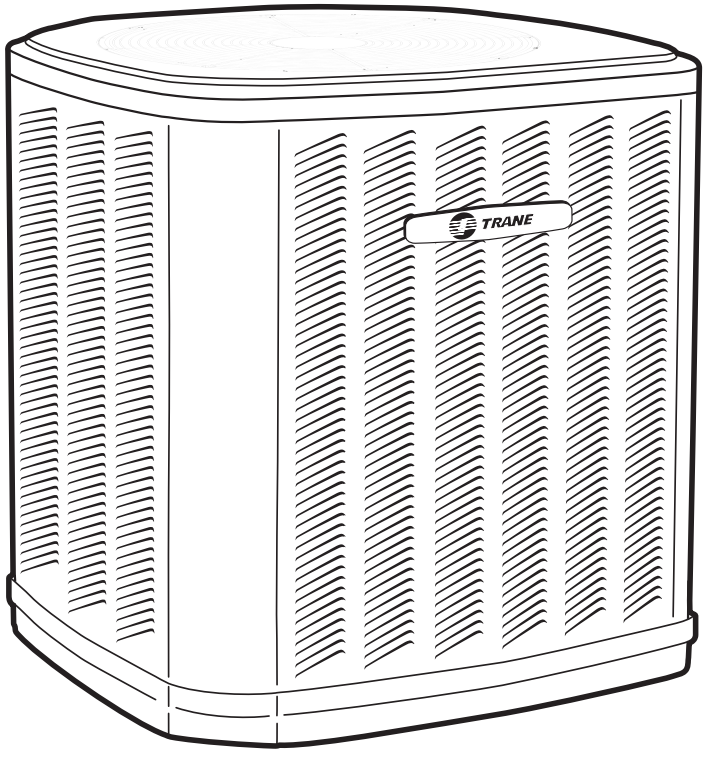
TRANE®

Split System Cooling Product Data

4TTB3018-060

1½ – 5 Tons



PUB. NO. 22-1843-10

Features and Benefits

- Efficiency up to 13.25 SEER
- All aluminum **Spine Fin™** coil
- **WeatherGuard™** fasteners
- **Quick-Sess™** cabinet, easy service access and refrigerant connections with full coil protection
- Fast complete drain, weather proof base
- **Comfort "R"™** mode approved
- Glossy corrosion resistant finish
- Internal high/low pressure and temperature protection
- 018, 024, 030 ship with start kit
- Liquid line filter-drier
- Polyslate gray cabinet with anthracite gray badge & cap
- High pressure switch
- R-410A refrigerant
- S.E.E.T. design testing
- 100% line run test
- Low ambient cooling to 55°F as shipped
- Low ambient cooling to 30°F with AY28X079
- Low ambient cooling to 0°F with BAYLOAM103

- **Extended warranties available**

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- Accessory Description and Usage 5
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General Data

Product Specifications

Model No. ①	4TTB3018E1	4TTB3024E1	4TTB3030E1	4TTB3036E1
Electrical Data V/Ph/Hz ②	208/230/1/60	208/230/1/60	200/230/1/60	208/230/1/60
Min Branch Cir Ampacity	9	12	15	20
Br. Cir. Prot. Rtg. - Max (Amps)	15	20	25	35
Compressor	SCROLL	SCROLL	SCROLL	SCROLL
RL Amps - LR Amps	6.4 - 40	8.9 - 48.5	11.5 - 63.5	15.3 - 70
Outdoor Fan FL Amps	0.74	0.74	0.74	0.74
Fan HP	1/8	1/8	1/8	1/8
Fan Dia (inches)	23.0	23.0	23.0	23.0
Coil	Spine Fin™	Spine Fin™	Spine Fin™	Spine Fin™
Refrigerant R-410A	4/15-LB/OZ	5/8-LB/OZ	5/2-LB/OZ	5/12-LB/OZ
Line Size - (in.) O.D. Gas ③	5/8	5/8	3/4	3/4
Line Size - (in.) O.D. Liquid ③	3/8	3/8	3/8	3/8
Charge Spec. Subcooling	8°	10°	10°	10°
Dimensions H x W x D (Crated)	34 x 30.1 x 33	34 x 30.1 x 33	34 x 30.1 x 33	34 x 30.1 x 33
Weight - Shipping	191	196	200	186
Weight - Net	165	169	173	159
Start Components	YES	YES	YES	NO
Sound Enclosure	YES	YES	NO	NO
Compressor Sump Heat	NO	NO	NO	NO
Optional Accessories: ④				
Anti-short Cycle Timer	TAYASCT501A	TAYASCT501A	TAYASCT501A	TAYASCT501A
Evaporator Defrost Control	AY28X079	AY28X079	AY28X079	AY28X079
Rubber Isolator Kit	BAYISLT101	BAYISLT101	BAYISLT101	BAYISLT101
Extreme Condition Mounting Kit	BAYECMT023	BAYECMT023	BAYECMT023	BAYECMT023
Start Kit				BAYKSKT263
Crankcase Heater Kit	BAYCCHT300	BAYCCHT300	BAYCCHT300	BAYCCHT301
Seacoast Kit	BAYSEAC001	BAYSEAC001	BAYSEAC001	BAYSEAC001
Low Ambient Kit	BAYLOAM103	BAYLOAM103	BAYLOAM103	BAYLOAM103
Refrigerant Lineset ⑤	TAYREFLN950	TAYREFLN950	TAYREFLN7*	TAYREFLN7*

① Certified in accordance with the Unitary Air-Conditioner equipment certification program which is based on AHRI Standard 210/240.
② Calculated in accordance with N.E.C. Only use HACR circuit breakers or fuses.
③ Standard line lengths - 60'. Standard lift - 60' Suction and Liquid line.
For Greater lengths and lifts refer to refrigerant piping software Pub# 32-3312-0†. (†denotes latest revision)
④ For accessory description and usage, see page 5.
⑤ * = 15, 20, 25, 30, 40 and 50 foot lineset available.

MODEL	SOUND POWER LEVEL [dB(A)]	A-WEIGHTED FULL OCTAVE SOUND POWER LEVEL dB [(dB(A)]							
		63	125	250	500	1000	2000	4000	5000
4TTB3018E1	78	45.5	59.6	62.2	69.3	71.1	70.9	62.2	54.3
4TTB3024E1	78	47.9	60.5	64.1	71.2	71.2	69.0	58.2	51.5
4TTB3030E1	78	49.0	60.2	66.0	70.3	71.4	69.8	60.4	53.0
4TTB3036E1	74	47.5	64.5	62.3	69.1	71.2	72.6	59.9	52.1
4TTB3042D1	79	47.5	64.5	67.0	75.3	74.0	70.7	62.2	52.8
4TTB3048D1	79	47.4	60.0	66.9	75.3	73.5	70.3	62.0	51.4
4TTB3060D1	80	47.3	55.7	69.0	72.7	75.8	69.4	62.2	53.3
Note: Rated in accordance with AHRI Standard 270-2008									

General Data

Product Specifications

Model No. ①	4TTB3042D1	4TTB3048D1	4TTB3060D1
Electrical Data V/Ph/Hz ②	208/230/1/60	208/230/1/60	208/230/1/60
Min Cir Ampacity	26	28	35
Max Fuse Size (Amps)	45	50	60
Compressor	SCROLL	SCROLL	SCROLL
RL Amps - LR Amps	19.9 - 105	21.8 - 117	26.8 - 134
Outdoor Fan FL Amps	0.93	0.93	0.93
Fan HP	1/5	1/5	1/5
Fan Dia (inches)	27.5	27.5	27.5
Coil	Spine Fin™	Spine Fin™	Spine Fin™
Refrigerant R-410A	9/2-LB/OZ	6/13-LB/OZ	8/0-LB/OZ
Line Size - (in.) O.D. Gas ③	3/4	7/8	7/8
Line Size - (in.) O.D. Liquid ③	3/8	3/8	3/8
Charge Spec. Subcooling	10°	10°	10°
Dimensions H x W x D (Crated)	34.4 x 35.1 x 38.7	34.4 x 35.1 x 38.7	42.4 x 35.1 x 38.7
Weight - Shipping	228	235	261
Weight - Net	196	203	226
Start Components	NO	NO	NO
Sound Enclosure	NO	NO	NO
Compressor Sump Heat	NO	NO	NO
Optional Accessories: ④			
Anti-short Cycle Timer	TAYASCT501A	TAYASCT501A	TAYASCT501A
Evaporator Defrost Control	AY28X079	AY28X079	AY28X079
Rubber Isolator Kit	BAYISLT101	BAYISLT101	BAYISLT101
Extreme Condition Mounting Kit	BAYECMT004	BAYECMT004	BAYECMT004
Start Kit	BAYKSKT263	BAYKSKT263	BAYKSKT263
Crankcase Heater Kit	BAYCCHT301	BAYCCHT301	BAYCCHT301
Seacoast Kit	BAYSEAC001	BAYSEAC001	BAYSEAC001
Low Ambient Kit	BAYLOAM103	BAYLOAM103	BAYLOAM103
Refrigerant Lineset ⑤	TAYREFLN7*	TAYREFLN3*	TAYREFLN3*

Accessory Description and Usage

Anti-Short Cycle Timer — Solid state timing device that prevents compressor recycling until five (5) minutes have elapsed after satisfying call or power interruptions. Use in area with questionable power delivery, commercial applications, long lineset, etc.

Evaporator Defrost Control — SPST Temperature actuated switch that cycles the condenser off as indoor coil reaches freeze-up conditions. Used for low ambient cooling to 30°F with TXV.

Rubber Isolators — Five (5) large rubber donuts to isolate condensing unit from transmitting energy into mounting frame or pad. Use on any application where sound transmission needs to be minimized.

Hard Start kit — Start capacitor and relay to assist compressor motor startup. Use in areas with marginal power supply, on long linesets, low ambient conditions, etc.

Extreme Condition Mount Kit — Bracket kits to securely mount condensing unit to a frame or pad without removing any panels. Use in areas with high winds, or on commercial roof tops, etc.

AHRI Standard Capacity Rating Conditions

AHRI STANDARD 210/240 RATING CONDITIONS —
(A) Cooling 80°F DB, 67°F WB air entering indoor coil, 95°F DB air entering outdoor coil.
(B) High Temperature Heating 47°F DB, 43°F WB air entering outdoor coil, 70°F DB air entering indoor coil.
(C) Low Temperature Heating 17°F DB, 15°F WB air entering outdoor coil, 70°F DB air entering indoor coil.
(D) Rated indoor airflow for heating is the same as for cooling.

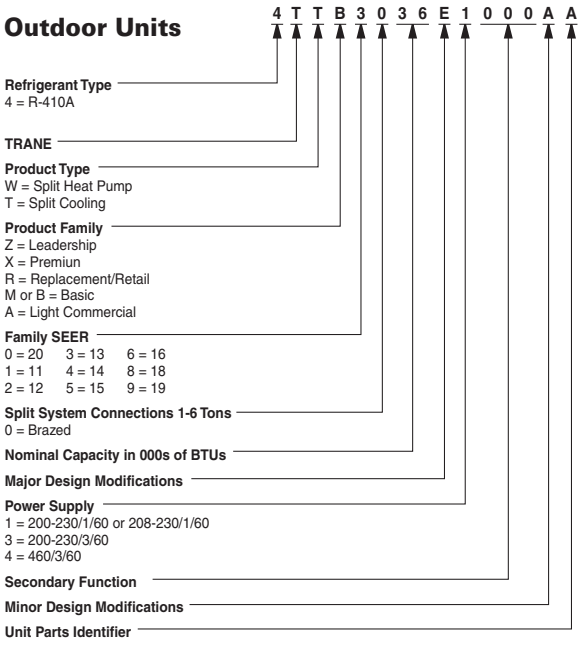
AHRI STANDARD 270 RATING CONDITIONS — (Noise rating numbers are determined with the unit in cooling operation.) Standard Noise Rating number is at 95°F outdoor air.



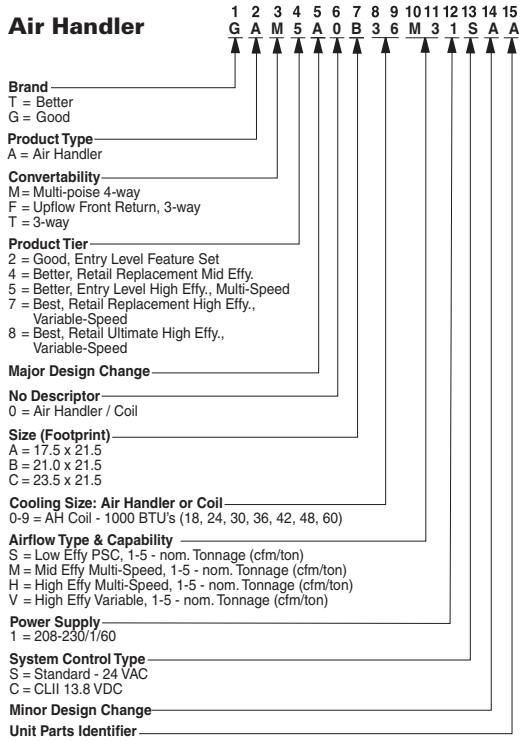


Model Nomenclature

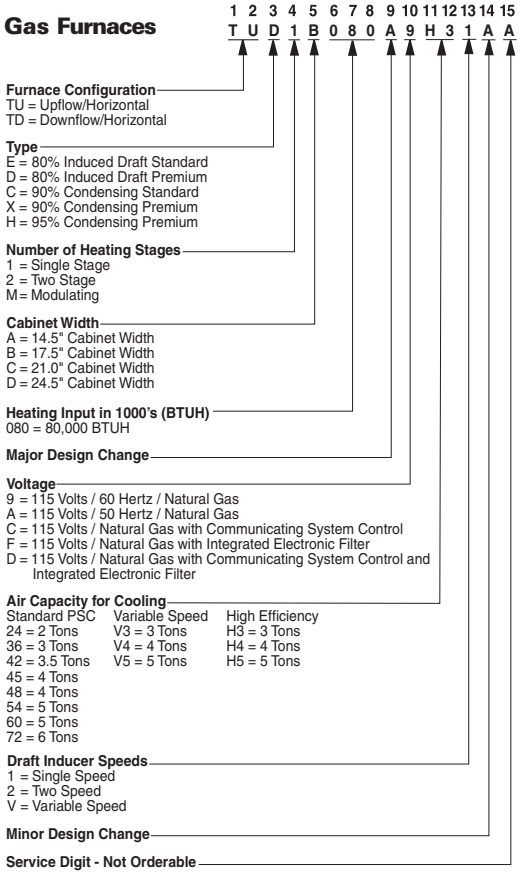
Outdoor Units



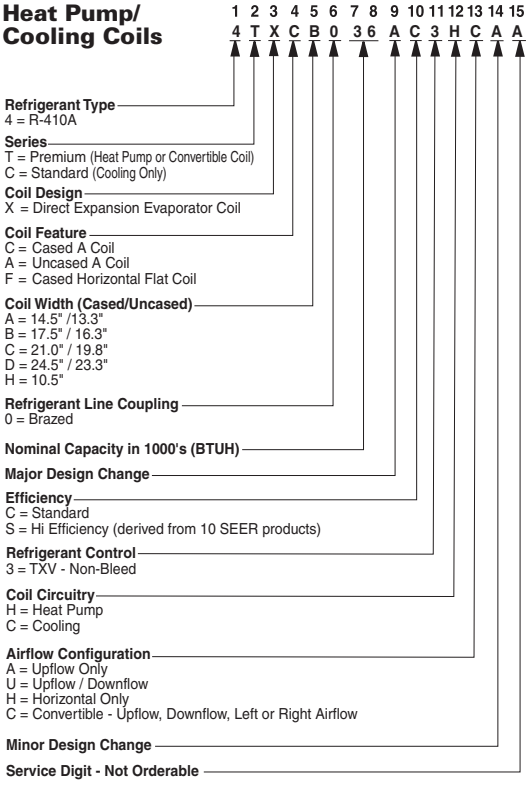
Air Handler



Gas Furnaces



Heat Pump/ Cooling Coils

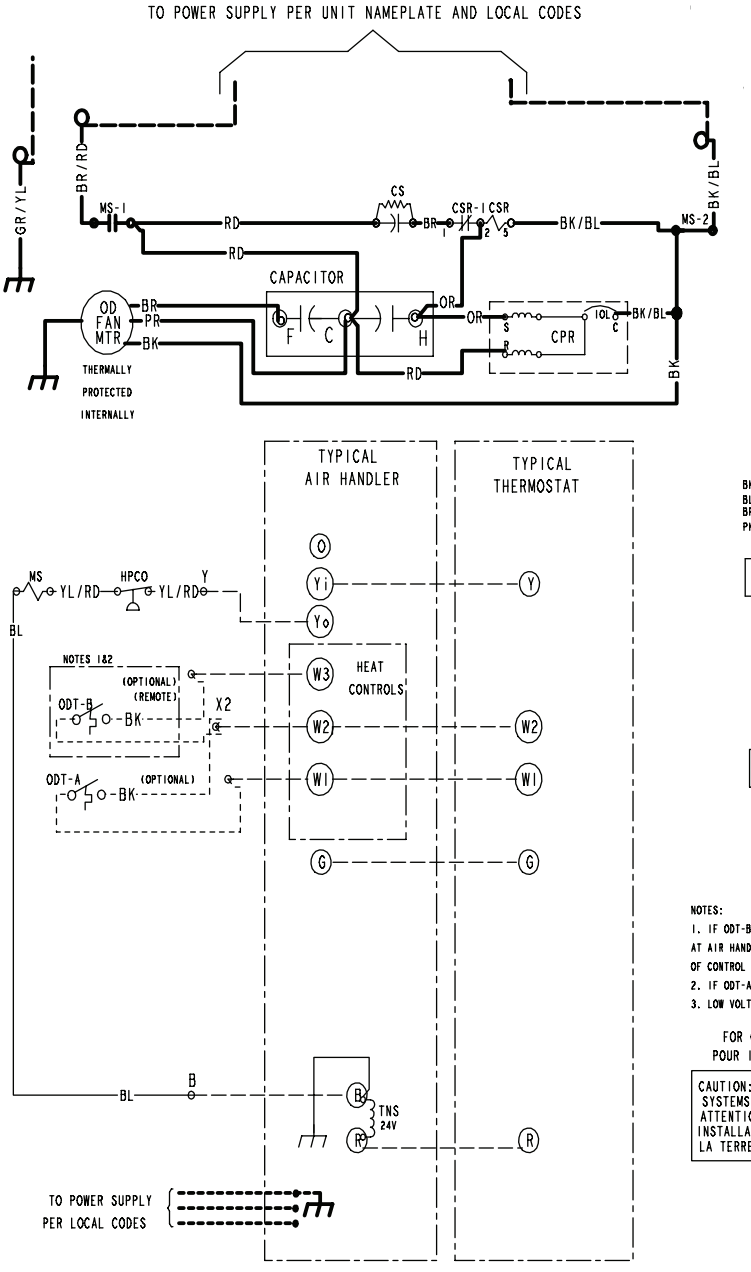


Electrical Data

SCHEMATIC DIAGRAMS

(SEE LEGEND)

4TTB3018,024,030E1



CF	FAN CAPACITOR
CN	WIRECONNECTOR
CPR	COMPRESSOR
CR	RUN CAPACITOR
CS	STARTING CAPACITOR
CSR	CAPACITOR SWITCHING RELAY
F	INDOOR FAN RELAY
HPCO	HIGH PRESSURE CUTOUT SWITCH
LPCO	LOW PRESSURE CUTOUT SWITCH
IOL	INTERNAL OVERLOAD PROTECTOR
SM	SYSTEM ON-OFF SWITCH
MS	COMPRESSOR MOTOR CONTACTOR
ODA	OUTDOOR ANTICIPATOR
OFT	OUTDOOR FAN THERMOSTAT
ODS	OUTDOOR TEMPERATURE SENSOR
ODT	OUTDOOR THERMOSTAT
SC	SWITCH OVER VALVE SOLENOID
TDL	DISCHARGE LINE THERMOSTAT
TNS	TRANSFORMER

COLOR OF WIRE
BK/BL
COLOR OF MARKER

BK	BLACK	RD	RED
BL	BLUE	WH	WHITE
BR	BROWN	YL	YELLOW
PK	PINK	OR	ORANGE
		GR	GREEN
		PR	PURPLE

⚠WARNING

HAZARDOUS VOLTAGE!

DISCONNECT ALL ELECTRICAL POWER INCLUDING REMOTE DISCONNECTS BEFORE SERVICING.

Failure to disconnect power before servicing can cause severe personal injury or death.

⚠CAUTION

USE COPPER CONDUCTORS ONLY!

UNIT TERMINALS ARE NOT DESIGNED TO ACCEPT OTHER TYPES OF CONDUCTORS.

Failure to do so may cause damage to the equipment.

NOTES:

1. IF ODT-B IS NOT USED. ADD JUMPER BETWEEN W2 & W3 AT AIR HANDLER. IF USED, ODT-B MUST BE MOUNTED REMOTE OF CONTROL BOX IN AN APPROVED WEATHER PROOF ENCLOSURE.
2. IF ODT-A IS NOT USED. ADD JUMPER BETWEEN W1 & W2 AT AIR HANDLER.
3. LOW VOLTAGE (24 V) FIELD WIRING MUST BE 18 AWG MIN.

FOR CANADIAN INSTALLATIONS
POUR INSTALLATIONS CANADIENNES

CAUTION: NOT SUITABLE FOR USE ON SYSTEMS EXCEEDING 150V-TO-GROUND
ATTENTION:NE CONVIENT PAS AUX INSTALLATIONS DE PLUS DE 150 V A LA TERRE

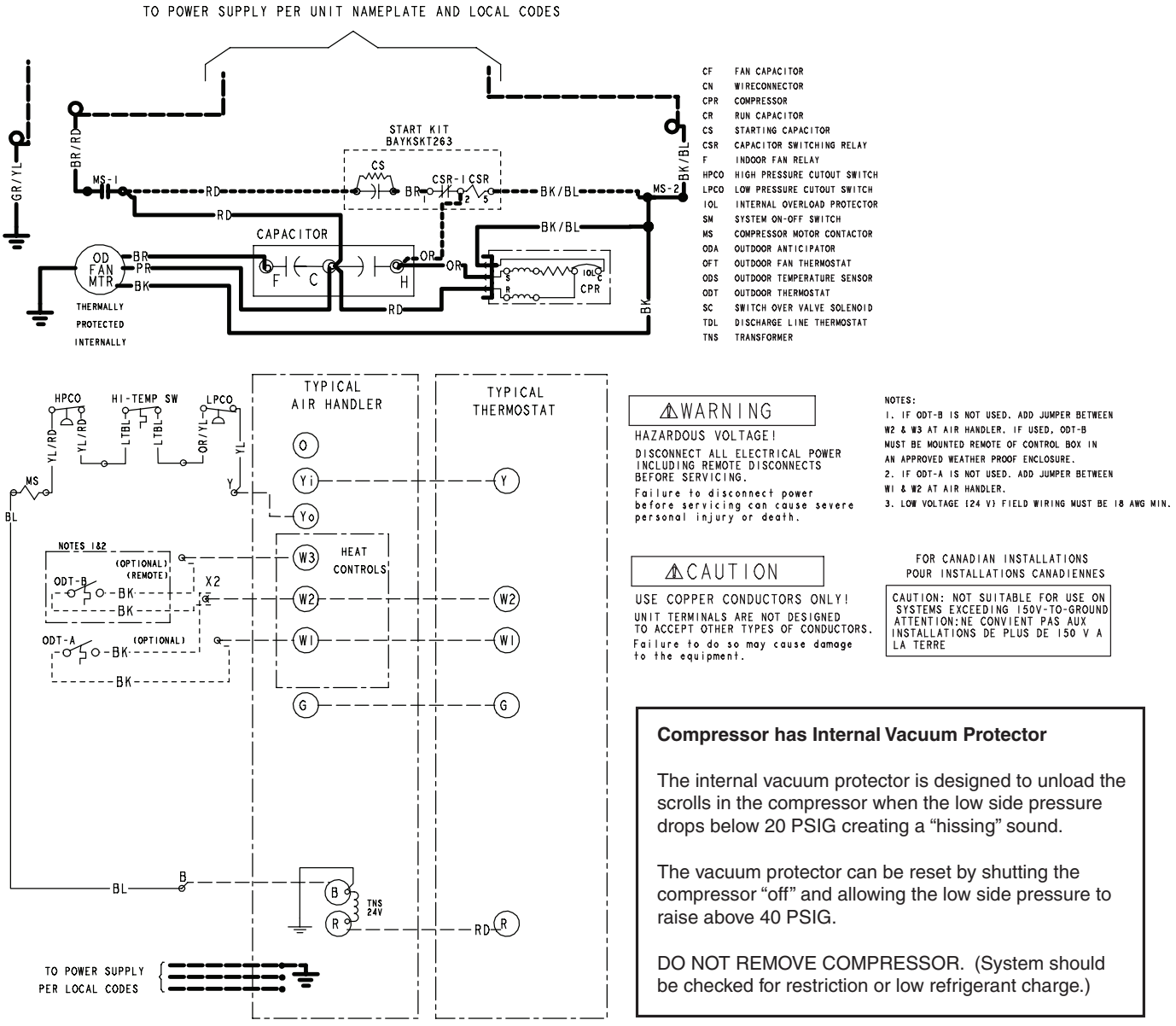
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Electrical Data

SCHEMATIC DIAGRAMS

(SEE LEGEND)

4TTB3036E1



⚠WARNING

HAZARDOUS VOLTAGE!

DISCONNECT ALL ELECTRICAL POWER INCLUDING REMOTE DISCONNECTS BEFORE SERVICING.

Failure to disconnect power before servicing can cause severe personal injury or death.

NOTES:

1. IF ODT-B IS NOT USED. ADD JUMPER BETWEEN W2 & W3 AT AIR HANDLER. IF USED, ODT-B MUST BE MOUNTED REMOTE OF CONTROL BOX IN AN APPROVED WEATHER PROOF ENCLOSURE.
2. IF ODT-A IS NOT USED. ADD JUMPER BETWEEN W1 & W2 AT AIR HANDLER.
3. LOW VOLTAGE (24 V) FIELD WIRING MUST BE 18 AWG MIN.

⚠CAUTION

USE COPPER CONDUCTORS ONLY!

UNIT TERMINALS ARE NOT DESIGNED TO ACCEPT OTHER TYPES OF CONDUCTORS.

Failure to do so may cause damage to the equipment.

FOR CANADIAN INSTALLATIONS
POUR INSTALLATIONS CANADIENNES

CAUTION: NOT SUITABLE FOR USE ON SYSTEMS EXCEEDING 150V-TO-GROUND
ATTENTION: NE CONVIENT PAS AUX INSTALLATIONS DE PLUS DE 150 V A LA TERRE

Compressor has Internal Vacuum Protector

The internal vacuum protector is designed to unload the scrolls in the compressor when the low side pressure drops below 20 PSIG creating a “hissing” sound.

The vacuum protector can be reset by shutting the compressor “off” and allowing the low side pressure to raise above 40 PSIG.

DO NOT REMOVE COMPRESSOR. (System should be checked for restriction or low refrigerant charge.)

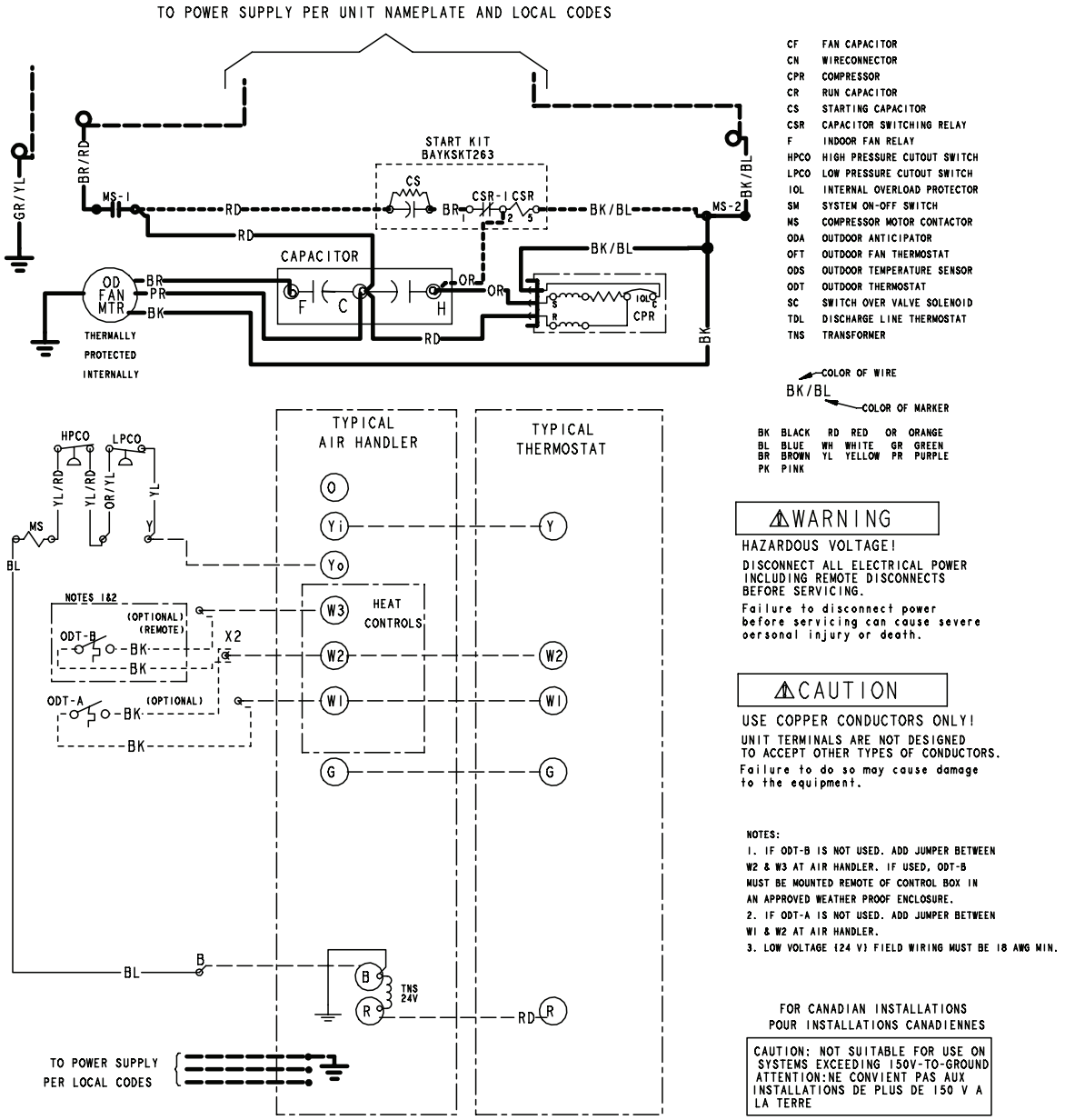
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Electrical Data

SCHEMATIC DIAGRAMS

(SEE LEGEND)

4TTB3042D1



CF	FAN CAPACITOR
CN	WIRECONNECTOR
CPR	COMPRESSOR
CR	RUN CAPACITOR
CS	STARTING CAPACITOR
CSR	CAPACITOR SWITCHING RELAY
F	INDOOR FAN RELAY
HPCO	HIGH PRESSURE CUTOUT SWITCH
LPCO	LOW PRESSURE CUTOUT SWITCH
IOL	INTERNAL OVERLOAD PROTECTOR
SM	SYSTEM ON-OFF SWITCH
MS	COMPRESSOR MOTOR CONTACTOR
ODA	OUTDOOR ANTICIPATOR
OFT	OUTDOOR FAN THERMOSTAT
ODS	OUTDOOR TEMPERATURE SENSOR
ODT	OUTDOOR THERMOSTAT
SC	SWITCH OVER VALVE SOLENOID
TDL	DISCHARGE LINE THERMOSTAT
TNS	TRANSFORMER

COLOR OF WIRE
BK/BL
COLOR OF MARKER

BK	BLACK	RD	RED	OR	ORANGE
BL	BLUE	WH	WHITE	GR	GREEN
BR	BROWN	YL	YELLOW	PR	PURPLE
PK	PINK				

⚠WARNING

HAZARDOUS VOLTAGE!

DISCONNECT ALL ELECTRICAL POWER INCLUDING REMOTE DISCONNECTS BEFORE SERVICING.

Failure to disconnect power before servicing can cause severe oersonal injury or death.

⚠CAUTION

USE COPPER CONDUCTORS ONLY!

UNIT TERMINALS ARE NOT DESIGNED TO ACCEPT OTHER TYPES OF CONDUCTORS.

Failure to do so may cause damage to the equipment.

NOTES:

1. IF ODT-B IS NOT USED. ADD JUMPER BETWEEN W2 & W3 AT AIR HANDLER. IF USED, ODT-B MUST BE MOUNTED REMOTE OF CONTROL BOX IN AN APPROVED WEATHER PROOF ENCLOSURE.
2. IF ODT-A IS NOT USED. ADD JUMPER BETWEEN W1 & W2 AT AIR HANDLER.
3. LOW VOLTAGE (24 V) FIELD WIRING MUST BE 18 AWG MIN.

FOR CANADIAN INSTALLATIONS
POUR INSTALLATIONS CANADIENNES

CAUTION: NOT SUITABLE FOR USE ON SYSTEMS EXCEEDING 150V-TO-GROUND
ATTENTION:NE CONVIENT PAS AUX INSTALLATIONS DE PLUS DE 150 V A LA TERRE

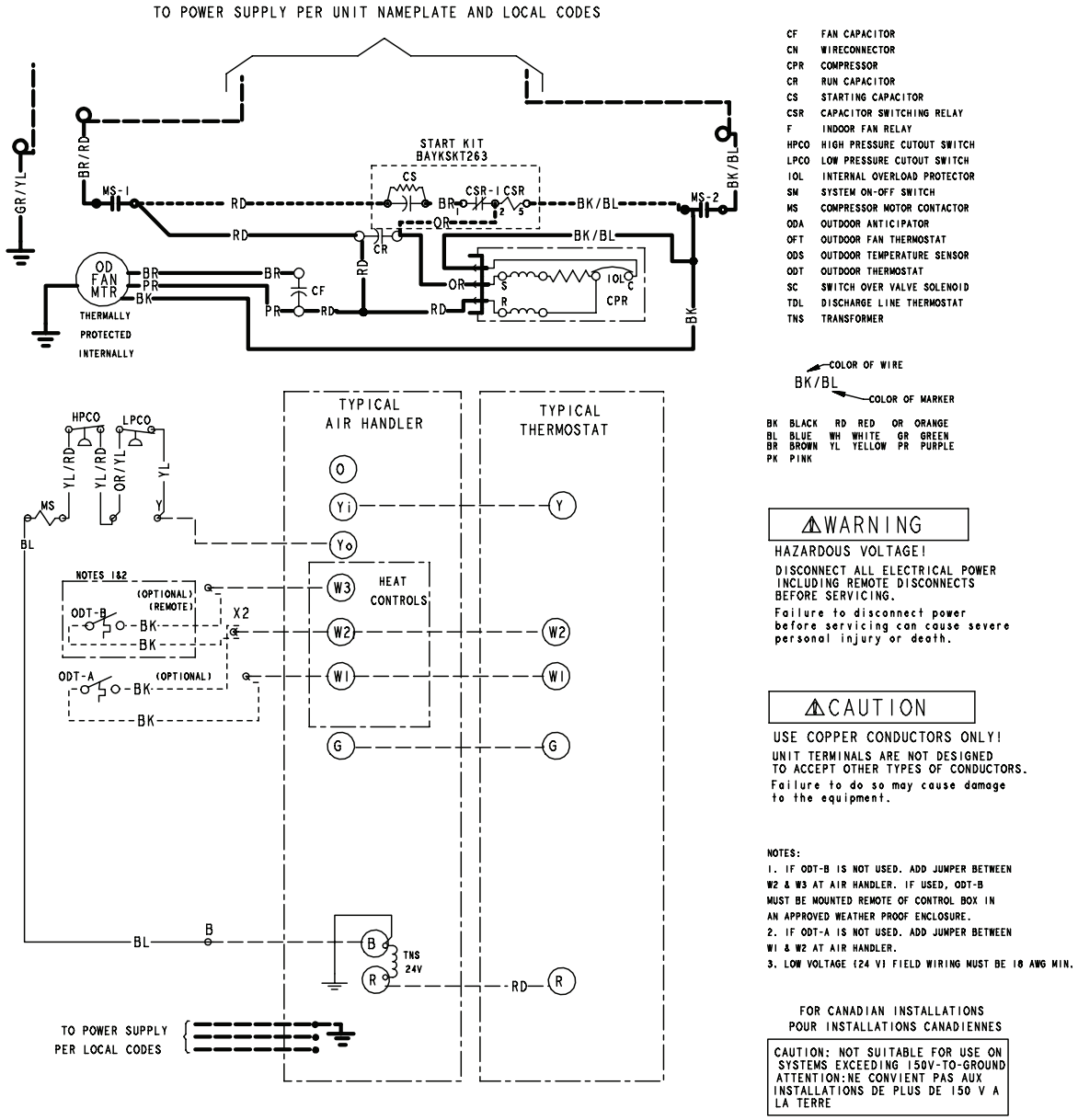
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Electrical Data

SCHEMATIC DIAGRAMS

(SEE LEGEND)

4TTB3048 and 060D1



CF FAN CAPACITOR
CN WIRECONNECTOR
CPR COMPRESSOR
CR RUN CAPACITOR
CS STARTING CAPACITOR
CSR CAPACITOR SWITCHING RELAY
F INDOOR FAN RELAY
HPCO HIGH PRESSURE CUTOUT SWITCH
LPCO LOW PRESSURE CUTOUT SWITCH
IOL INTERNAL OVERLOAD PROTECTOR
SM SYSTEM ON-OFF SWITCH
MS COMPRESSOR MOTOR CONTACTOR
ODA OUTDOOR ANTICIPATOR
OFT OUTDOOR FAN THERMOSTAT
ODS OUTDOOR TEMPERATURE SENSOR
ODT OUTDOOR THERMOSTAT
SC SWITCH OVER VALVE SOLENOID
TDL DISCHARGE LINE THERMOSTAT
TNS TRANSFORMER

COLOR OF WIRE
BK/BL
COLOR OF MARKER

BK BLACK RD RED OR ORANGE
BL BLUE WH WHITE GR GREEN
BR BROWN YL YELLOW PR PURPLE
PK PINK

⚠WARNING

HAZARDOUS VOLTAGE!

DISCONNECT ALL ELECTRICAL POWER INCLUDING REMOTE DISCONNECTS BEFORE SERVICING.

Failure to disconnect power before servicing can cause severe personal injury or death.

⚠CAUTION

USE COPPER CONDUCTORS ONLY!

UNIT TERMINALS ARE NOT DESIGNED TO ACCEPT OTHER TYPES OF CONDUCTORS.

Failure to do so may cause damage to the equipment.

NOTES:
1. IF ODT-B IS NOT USED. ADD JUMPER BETWEEN W2 & W3 AT AIR HANDLER. IF USED, ODT-B MUST BE MOUNTED REMOTE OF CONTROL BOX IN AN APPROVED WEATHER PROOF ENCLOSURE.
2. IF ODT-A IS NOT USED. ADD JUMPER BETWEEN W1 & W2 AT AIR HANDLER.
3. LOW VOLTAGE (24 V) FIELD WIRING MUST BE 18 AWG MIN.

FOR CANADIAN INSTALLATIONS
POUR INSTALLATIONS CANADIENNES

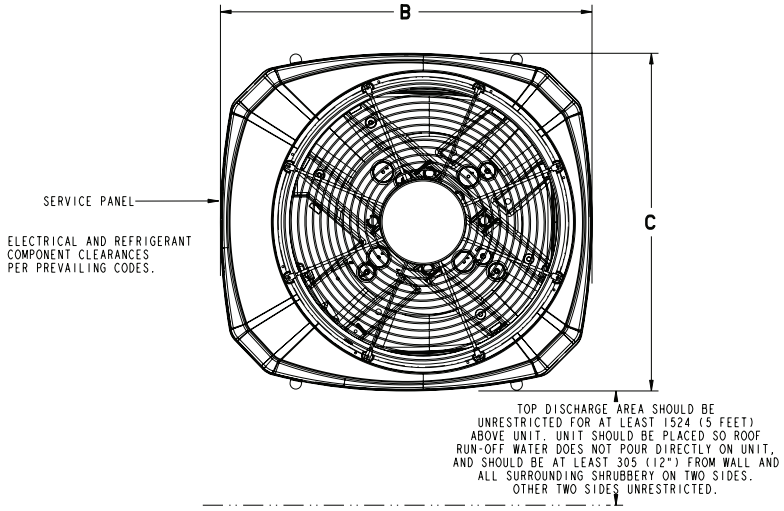
CAUTION: NOT SUITABLE FOR USE ON SYSTEMS EXCEEDING 150V-TO-GROUND
ATTENTION:NE CONVIENT PAS AUX INSTALLATIONS DE PLUS DE 150 V A LA TERRE

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Dimensions

4TTB3 OUTLINE DRAWING

NOTE: ALL DIMENSIONS ARE IN MM (INCHES)



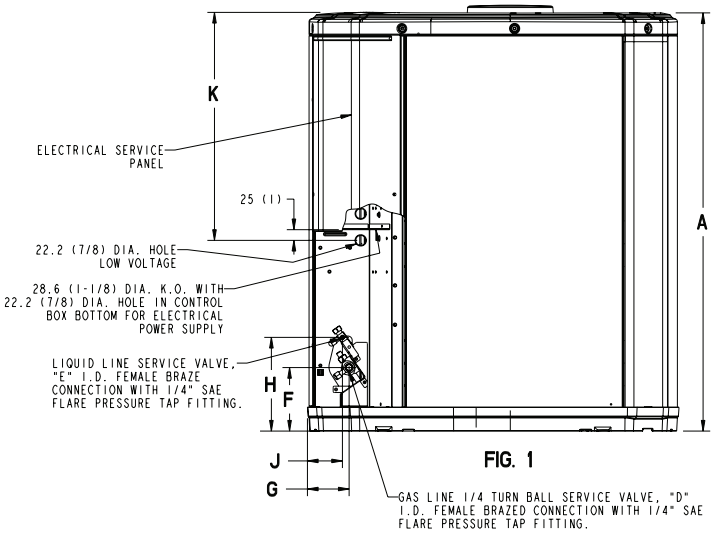


FIG. 1

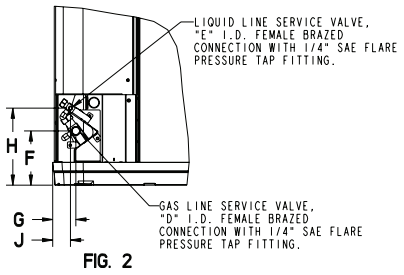


FIG. 2

MODELS	BASE	FIG.	A	B	C	D	E	F	G	H	J	K
4TTB3018E1	3	1	730 (28-3/4)	829 (32-5/8)	756 (29-3/4)	5/8	3/8	143 (5-5/8)	92 (3-5/8)	210 (8-1/4)	79 (3-1/8)	508 (20)
4TTB3024E1	3	1	730 (28-3/4)	829 (32-5/8)	756 (29-3/4)	5/8	3/8	143 (5-5/8)	92 (3-5/8)	210 (8-1/4)	79 (3-1/8)	508 (20)
4TTB3030E1	3	1	730 (28-3/4)	829 (32-5/8)	756 (29-3/4)	3/4	3/8	143 (5-5/8)	92 (3-5/8)	210 (8-1/4)	79 (3-1/8)	508 (20)
4TTB3036E1	3	1	730 (28-3/4)	829 (32-5/8)	756 (29-3/4)	3/4	3/8	137 (5-3/8)	79 (3-1/8)	197 (7-3/4)	60 (2-3/8)	508 (20)
4TTB3042D1	4	1	741 (29-1/8)	946 (37-1/4)	870 (34-1/4)	3/4	3/8	152 (6)	98 (3-7/8)	219 (8-5/8)	86 (3-3/8)	508 (20)
4TTB3048D1	4	1	741 (29-1/8)	946 (37-1/4)	870 (34-1/4)	7/8	3/8	152 (6)	98 (3-7/8)	219 (8-5/8)	86 (3-3/8)	508 (20)
4TTB3060D1	4	1	943 (37-1/8)	946 (37-1/4)	870 (34-1/4)	7/8	3/8	152 (6)	98 (3-7/8)	219 (8-5/8)	86 (3-3/8)	508 (20)

From Dwg. D153074

Mechanical Specification Options

General
The 4TTB3 models shall be fully charged from the factory for matched indoor section and up to 15 feet of piping. This unit must be designed to operate at outdoor ambient temperatures as high as 115°F. Cooling capacities shall be matched with a wide selection of air handlers and furnace coils that are AHRI certified. The unit is certified to UL 1995. Exterior must be designed for outdoor application.

Casing
Unit casing is constructed of heavy gauge, galvanized steel and painted with a weather-resistant powder paint. Corrosion and weatherproof CMBP-G30 base.

Refrigerant Controls
Refrigeration system controls include condenser fan and compressor contactor. High and low pressure controls are inherent to the compressor. Another standard feature is the liquid line dryer.

Compressor
The compressor features internal over temperature and pressure protector, total dipped hermetic motor and thermostatically controlled sump heater. Other features include: Centrifugal oil pump, and low vibration and noise.

Condenser Coil
The Spine Fin™ coil shall be continuously wrapped, corrosion resistant all aluminum with minimum brazed joints. This coil is 3/8 inch O.D. seamless aluminum glued to a continuous aluminum fin. Coils are lab tested to withstand 2,000 pounds of pressure per square inch. The outdoor coil provides low airflow resistance and efficient heat transfer. The coil is protected on all four sides by louvered panels.

Low Ambient Cooling
As manufactured, this unit has a cooling capability to 55°F. The addition of an evaporator defrost control permits operation to 30°F. The addition of a low ambient kit with TXV permits low ambient cooling to 0°F.

Accessories
Thermostats — Heating/Cooling (manual and automatic changeover). Sub-base to match thermostat and locking thermostat cover.

Evaporator Defrost Control — See Low Ambient Cooling.

Outdoor Thermostat — Supplemental heat outdoor ambient lockout from 46 to –10°F.





TRANE®

Trane
www.trane.com

05/12

*Trane has a policy of continuous product and product data improvement **and** it reserves the right to change design and specifications without notice.*